```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import time

from sklearn.manifold import TSNE, LocallyLinearEmbedding

%matplotlib inline

train_df = pd.read_csv('mnist_train.csv')

y = train_df['label'].values                # the digit (0-9)
X = train_df.drop('label', axis=1).values       # the 784 pixel columns
X = X / 255.0                           # normalize pixels to [0, 1]

print("Images  :", X.shape[0])
print("Features:", X.shape[1])
print("Classes :", len(np.unique(y)))

start = time.time()
tsne_emb = TSNE(n_components=2, perplexity=30, random_state=42).fit_transform(X)
print(f"t-SNE time: {time.time() - start:.1f} s")

plt.figure(figsize=(8, 6))
plt.scatter(tsne_emb[:, 0], tsne_emb[:, 1], c=y, cmap='tab10', s=8)
plt.colorbar(label='Digit')
```

```
plt.title('t-SNE (perplexity = 30)')
plt.show()

plt.figure(figsize=(14, 12))

for i, p in enumerate([5, 30, 50, 100], 1):
    emb = TSNE(n_components=2, perplexity=p, random_state=42).fit_transform(X)
    plt.subplot(2, 2, i)
    plt.scatter(emb[:, 0], emb[:, 1], c=y, cmap='tab10', s=8)
    plt.colorbar(label='Digit')
    plt.title(f'Perplexity = {p}')

plt.show()

start = time.time()
lle_emb = LocallyLinearEmbedding(n_neighbors=10, n_components=2,
random_state=42).fit_transform(X)
print(f"LLE time: {time.time() - start:.1f} s")

plt.figure(figsize=(8, 6))
plt.scatter(lle_emb[:, 0], lle_emb[:, 1], c=y, cmap='tab10', s=8)
plt.colorbar(label='Digit')
plt.title('LLE (n_neighbors = 10)')
plt.show()

plt.figure(figsize=(14, 12))
```

```
for i, k in enumerate([5, 10, 30, 50], 1):
    emb = LocallyLinearEmbedding(n_neighbors=k, n_components=2,
random_state=42).fit_transform(X)
    plt.subplot(2, 2, i)
    plt.scatter(emb[:, 0], emb[:, 1], c=y, cmap='tab10', s=8)
    plt.colorbar(label='Digit')
    plt.title(f'n_neighbors = {k}')

plt.show()



plt.figure(figsize=(16, 6))

plt.subplot(1, 2, 1)
plt.scatter(tsne_emb[:, 0], tsne_emb[:, 1], c=y, cmap='tab10', s=8)
plt.colorbar(label='Digit')
plt.title('t-SNE (perplexity = 30)')

plt.subplot(1, 2, 2)
plt.scatter(lle_emb[:, 0], lle_emb[:, 1], c=y, cmap='tab10', s=8)
plt.colorbar(label='Digit')
plt.title('LLE (n_neighbors = 10)')

plt.show()
```